

Course Code- 151604

AI and Ethics

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Unit-1.0:**2 hrs****Introduction:** Definition of morality and ethics in AI, AI, and society.**Unit-2.0:****2 hrs****Impact of AI:** Impact on society, Impact on human psychology, Impact on the legal system, Impact on the environment and the planet, and Impact on trust.**Unit-3.0:****2 hrs****Ethical Initiatives in AI:** International ethical initiatives, Ethical harms, and concerns, Case study: healthcare robots, Autonomous Vehicles, Warfare, and weaponization.**Unit-4.0:****4 hrs****AI Standards And Regulation:** Model Process for Addressing Ethical Concerns During System Design, Transparency of Autonomous Systems, Data Privacy Process, Algorithmic Bias Considerations, Ontological Standard for Ethically Driven Robotics and Automation Systems.**Unit-5.0:****Rob ethics: Social and Ethical Implication of Robotics:****2 hrs**

Robotics- Ethics and Morality, Moral Theories-Ethics in Science and Technology, Ethical Issues in an ICT Society, Harmonization of Principles, Ethics and Professional Responsibility, Robotics Taxonomy.

Unit-6.0:**2 hrs****Challenges and Opportunities:** Ethical issues in artificial intelligence, Societal Issues Concerning the Application of Artificial Intelligence in Medicine, decision-making role in industries, and National and International Strategies on AI.**Text/Reference:**

1. Eleanor Bird, Jasmin Fox-Skelly, Nicola Jenner, Ruth Larbey, Emma Weitkamp and Alan Winfield,” The ethics of artificial intelligence: Issues and initiatives”, EPRS | European Parliamentary Research Service Scientific Foresight Unit (STOA) PE 634.452– March 2020.
2. Patrick Lin, Keith Abney, George A Bekey,” Robot Ethics: The Ethical and Social Implications of Robotics”, The MIT Press- January 2014.
3. Towards a Code of Ethics for Artificial Intelligence (Artificial Intelligence: Foundations, Theory, and Algorithms) by Paula Boddington, November 2017.
4. Mark Coeckelbergh,” AI Ethics”, The MIT Press Essential Knowledge series, April 2020.